

Alignment of a Localized Cartan with the Bernoulli Core: A Reduction to a Mixed Normalizer Cocycle

July 2026

Abstract

Let $M = A \rtimes_\sigma G$, where A is a diffuse II_1 factor, and let $Q \subset M$ be Cartan. We prove that the intertwining relation $Q \prec_M A$ already implies that Q is unitarily conjugate to a Cartan subalgebra $D \subset A$. Fourier analysis then shows that D is Cartan in A , and that every translate $\sigma_g(D)$ is unitarily conjugate to D inside A . Thus strong localization gives considerably more than a finite bimodule.

Exact invariance is reduced to a concrete nonabelian cocycle problem with values in the normalizer of the hyperfinite Cartan pair. For $G = \Gamma \times \Gamma$ with Γ free, the obstruction can be made trivial on each factor separately, leaving a single mixed commutator cocycle. We formulate the precise vanishing theorem that would finish the alignment implication and outline a deformation strategy using the Jones projection of D .

1 Spatially moving the Cartan into the core

Let $A \subset M$ be an inclusion of finite factors and let $Q \subset M$ be Cartan.

Lemma 1.1 (One-column reduction). *If $Q \prec_M A$, then there are nonzero projections $e \in Q$, $f \in A$, and a partial isometry $v \in M$ such that*

$$v^*v = e, \quad vv^* = f, \quad v(eQ)v^* \subset fAf.$$

Proof. Popa intertwining gives a nonzero Q - A subbimodule

$$\mathcal{H} \subset L^2(M)$$

of finite right A -dimension. Since Q is diffuse abelian, cut $\mathcal{H}$ by a nonzero projection $e \in Q$ so that

$$\dim(e\mathcal{H}_A) \leq 1.$$

A right A -module of dimension at most one is isomorphic to $fL^2(A)$ for a projection $f \in A$. The inclusion of this module into $L^2(M)$ is implemented by a partial isometry $v \in M$. The left eQe -action is carried to left multiplication by fAf , giving the stated relations. $\square$

Theorem 1.2 (Spatial core alignment). *Assume that A is a diffuse II_1 factor. If $Q \subset M$ is Cartan and $Q \prec_M A$, then there is a unitary $u \in M$ such that*

$$D := uQu^* \subset A.$$

Proof. Take e, f, v from Lemma 1.1. Because Q is Cartan in the factor M , its Weyl equivalence relation is ergodic. Hence there is a countable family of partial isometries

$$n_i \in \mathcal{GN}_M(Q)$$

such that

$$e_i := n_i^* n_i \leq e, \quad r_i := n_i n_i^* \in Q, \quad r_i r_j = 0 \ (i \neq j), \quad \sum_i r_i = 1.$$

Put

$$f_i = v e_i v^* \in A.$$

Then

$$\sum_i \tau(f_i) = \sum_i \tau(e_i) = \sum_i \tau(r_i) = 1.$$

Choose pairwise orthogonal projections $p_i \in A$ such that

$$\sum_i p_i = 1, \quad \tau(p_i) = \tau(f_i),$$

and partial isometries $b_i \in A$ with

$$b_i^* b_i = f_i, \quad b_i b_i^* = p_i.$$

Define

$$u_i = b_i v n_i^*.$$

Then

$$u_i^* u_i = r_i, \quad u_i u_i^* = p_i.$$

Thus $u = \sum_i u_i$ converges strongly to a unitary in M .

For $x \in Q$, all off-diagonal terms vanish and

$$u x u^* = \sum_i b_i v (n_i^* x n_i) v^* b_i^* \in A.$$

Therefore $u Q u^* \subset A$. □

2 Fourier rigidity after spatial alignment

Let G be a countable group acting trace preservingly on A , and put

$$M = A \rtimes_\sigma G.$$

Assume that $D \subset A$ is maximal abelian in M and regular in M .

For $g \in G$, define

$$\mathcal{T}_g(D) = \left\{ w \in A : \begin{array}{l} w \text{ is a partial isometry,} \\ w w^* \in D, \quad w^* w \in \sigma_g(D), \\ w \sigma_g(D) w^* = D w w^* \end{array} \right\}.$$

Lemma 2.1 (Fourier coefficients of normalizers). *Let $v \in \mathcal{GN}_M(D)$, and write*

$$v = \sum_{g \in G} a_g u_g$$

in $L^2(M)$. Then the polar part of every nonzero a_g belongs to $\mathcal{T}_g(D)$.

Proof. Let α_v denote the partial automorphism of D implemented by v . The relation

$$vd = \alpha_v(d)v$$

gives, coefficientwise,

$$a_g \sigma_g(d) = \alpha_v(d) a_g, \quad d \in D.$$

Hence

$$a_g^* a_g \in \sigma_g(D)' \cap A = \sigma_g(D), \quad a_g a_g^* \in D' \cap A = D.$$

Taking the polar decomposition gives the conclusion. $\square$

Theorem 2.2 (Core Cartan and inner orbit rigidity). *Let $D \subset A$ be Cartan in $M = A \rtimes_\sigma G$. Then:*

1. D is Cartan in A ;
2. for every $g \in G$, there is a unitary $w_g \in A$ such that

$$w_g \sigma_g(D) w_g^* = D.$$

Proof. Let

$$B = \mathcal{N}_A(D)'.$$

For $v \in \mathcal{GN}_M(D)$, Lemma 2.1 applied at $g = e$ shows that

$$E_A(v) \in B.$$

Since the linear span of $\mathcal{GN}_M(D)$ is L^2 -dense in M ,

$$A = E_A(M) \subset B.$$

Thus $B = A$, proving that D is Cartan in A .

Fix $g \in G$. The coefficient map

$$\Phi_g : M \rightarrow A, \quad \Phi_g(x) = E_A(x u_g^*)$$

is L^2 -contractive and surjective. Since the span of $\mathcal{GN}_M(D)$ is dense in M , the linear span of the g -th Fourier coefficients of groupoid normalizers is dense in A . By Lemma 2.1, the linear span of $\mathcal{T}_g(D)$ is therefore dense in A .

Choose a maximal family $(w_i)_i \subset \mathcal{T}_g(D)$ whose range projections and source projections are respectively pairwise orthogonal. Put

$$r = 1 - \sum_i w_i w_i^*, \quad s = 1 - \sum_i w_i^* w_i.$$

The two projections have equal trace. If they were nonzero, density of the span of $\mathcal{T}_g(D)$ would give $t \in \mathcal{T}_g(D)$ with

$$rts \neq 0.$$

The polar part of rts would enlarge the maximal family, a contradiction. Therefore $r = s = 0$, and

$$w_g = \sum_i w_i$$

is a unitary satisfying

$$w_g \sigma_g(D) w_g^* = D.$$

$\square$

Corollary 2.3. *Let $Q \subset A \rtimes_\sigma G$ be Cartan. If $Q \prec_M A$, then there is a unitary $u \in M$ such that $D = u Q u^* \subset A$, D is Cartan in A , and every $\sigma_g(D)$ is unitarily conjugate to D inside A .*

3 The exact obstruction

Fix transporter unitaries

$$a_g \in \mathcal{U}(A), \quad a_g \sigma_g(D) a_g^* = D.$$

Their defect is

$$\Omega(g, h) = a_g \sigma_g(a_h) a_{gh}^*. \quad (1)$$

Both $a_g \sigma_g(a_h)$ and a_{gh} transport $\sigma_{gh}(D)$ onto D , so

$$\Omega(g, h) \in \mathcal{N}_A(D).$$

Changing a_g by left multiplication with an element of $\mathcal{N}_A(D)$ changes Ω by the corresponding nonabelian coboundary.

Proposition 3.1 (Exact alignment criterion). *The following are equivalent.*

1. *There is a unitary $z \in A$ such that*

$$\sigma_g(z D z^*) = z D z^* \quad \text{for every } g \in G.$$

2. *There are $z \in \mathcal{U}(A)$ and $n_g \in \mathcal{N}_A(D)$ such that*

$$z^* \sigma_g(z) = n_g a_g \quad \text{for every } g \in G. \quad (2)$$

Proof. The unitary $z^* \sigma_g(z)$ transports $\sigma_g(D)$ onto D exactly when $z D z^*$ is σ_g -invariant. Two transporters from $\sigma_g(D)$ to D differ on the left by an element of $\mathcal{N}_A(D)$. This is precisely (2). $\square$

Thus exact alignment is not ordinary $U(A)$ -valued cocycle triviality. It is a coboundary problem in the homogeneous space

$$\mathcal{N}_A(D) \backslash \mathcal{U}(A).$$

4 Reduction for two free factors

Now take

$$G = \Gamma_1 \times \Gamma_2, \quad \Gamma_i \cong \mathbb{F}_\infty.$$

Because each factor is free, transporter choices may be made multiplicative on each factor separately.

For the superrigidity statements below, identify both factors with Γ and specialize to the Krogager–Vaes action. Thus

$$A = \overline{\bigotimes_{k \in \Gamma} (A_0, \tau_0)}, \quad \theta : \Gamma \curvearrowright (A_0, \tau_0),$$

where (A_0, τ_0) is a nontrivial amenable finite tracial algebra and θ is trace preserving. Writing $\pi_k : A_0 \rightarrow A$ for the k -th tensor leg,

$$\sigma_{(g,h)}(\pi_k(x)) = \pi_{gkh^{-1}}(\theta_h(x)). \quad (3)$$

Proposition 4.1 (Mixed-cocycle reduction). *There are homomorphisms*

$$V : \Gamma_1 \rightarrow \mathcal{N}_M(D), \quad W : \Gamma_2 \rightarrow \mathcal{N}_M(D)$$

such that V_g has Fourier degree (g, e) and W_h has Fourier degree (e, h) . The only obstruction to combining them into a representation of G is

$$c(g, h) = V_g W_h V_g^* W_h^* \in \mathcal{N}_A(D).$$

Proof. Choose a homogeneous normalizing lift for each free generator and its inverse. Extend to reduced words. Freeness gives the homomorphisms V and W . Their commutator has degree e , so it lies in A ; since both terms normalize D , it lies in $\mathcal{N}_A(D)$. $\square$

For fixed h , the map $g \mapsto c(g, h)$ satisfies a nonabelian one-cocycle identity for the action $\text{Ad } V$ on $\mathcal{N}_A(D)$, and similarly in the other variable. The desired new theorem is:

Conjecture 4.2 (Mixed normalizer-cocycle vanishing). *For the Krogager–Vaes left–right Bernoulli action, the lifts in Proposition 4.1 can be changed by homogeneous elements of $\mathcal{N}_A(D)$ so that*

$$[V_g, W_h] = 0 \quad \text{for all } g, h.$$

The second coboundary assertion in the earlier formulation is not an additional conjecture. Once the two lifts commute, noncommutative Bernoulli cocycle superrigidity supplies it.

Proposition 4.3 (The H^1 finish). *Assume that the lifts in Proposition 4.1 have been perturbed inside $\mathcal{N}_A(D)$ so that their ranges commute. Then there is a unitary $z \in A$ such that*

$$\sigma_k(zDz^*) = zDz^* \quad \text{for every } k \in G.$$

Proof. The formula

$$U_{(g,h)} = V_g W_h$$

defines a representation of G in $\mathcal{N}_M(D)$. Write

$$U_k = b_k u_k, \quad b_k \in \mathcal{U}(A).$$

Multiplicativity gives

$$b_k \sigma_k(b_l) = b_{kl},$$

so $b : G \rightarrow \mathcal{U}(A)$ is a σ -cocycle. Also both $b_k u_k$ and $a_k u_k$ normalize D , and hence

$$b_k a_k^* \in \mathcal{N}_A(D).$$

We apply the noncommutative cocycle-superrigidity theorem [4, Theorem 4.1]. For completeness, its hypotheses are visible directly in the present action. Take $H = \Gamma \times \{e\}$ and $H' = \{e\} \times \Gamma$. The restriction to H is a generalized Bernoulli shift with stable spectral gap: its stabilizers on the coordinate set Γ are trivial. The restriction to H' has infinite coordinate orbits and acts weakly mixing. The usual free-product malleable deformation of the amenable base extends the internal action θ by the identity on the free $L(\mathbb{Z})$ -leg, and therefore commutes with the full left–right action. The subgroup H' is normal in G . The theorem, with scalar target algebra, provides $u \in \mathcal{U}(A)$ and a character $\chi : G \rightarrow \mathbb{T}$ such that

$$u^* b_k \sigma_k(u) = \chi(k).$$

Set $z = u^*$. Then

$$z^* \sigma_k(z) = u \sigma_k(u^*) = \chi(k)^* b_k = (\chi(k)^* b_k a_k^*) a_k.$$

The parenthesized unitary belongs to $\mathcal{N}_A(D)$. The exact alignment criterion, Proposition 3.1, finishes the proof. $\square$

Thus Conjecture 4.2, together with Corollary 2.3 and Proposition 4.3, proves the full alignment implication.

5 Graph projections and a deformation route

Let e_D be the Jones projection from $L^2(A)$ onto $L^2(D)$ and put

$$Z_D = D' \cap \langle A, e_D \rangle.$$

Since $D \subset A$ is Cartan, Z_D is the abelian relation algebra of its hyperfinite equivalence relation.

The left-multiplication formula $ve_Dv^* = e_D$ does *not* characterize normalizers; it usually characterizes the much smaller stabilizer of the subspace $L^2(D)$. The correct basic-construction formulation is the following.

Lemma 5.1 (Graph-projection criterion). *For $v \in \mathcal{U}(A)$,*

$$v \in \mathcal{N}_A(D) \iff ve_Dv^* \in Z_D.$$

Equivalently, on the standard form of A ,

$$(vJ_AvJ_A)e_D(vJ_AvJ_A)^* = e_D.$$

Proof. If v normalizes D , a direct computation shows that ve_Dv^* commutes with D . Conversely, if $ve_Dv^* \in D'$, then for every $d \in D$,

$$(v^*dv)e_D = e_D(v^*dv).$$

Applying both sides to the trace vector gives $v^*dv = E_D(v^*dv) \in D$. Hence $v^*Dv \subset D$, and equality follows from maximal abelianness. The standard-form statement is the usual implementation of the automorphism $\text{Ad } v$. $\square$

Thus the deformation must produce coherent *graph projections* in Z_D , rather than a left translate that fixes e_D . This suggests the following corrected route.

1. Enlarge the Bernoulli algebra to the usual malleable deformation $\tilde{A}$, and view e_D in the corresponding semifinite basic construction.
2. Use the two nonamenable lifted free-factor algebras generated by $V(\Gamma_1)$ and $W(\Gamma_2)$. Their mixed commutators lie in the amenable algebra A . Use hyperfinite finite subrelations to construct almost coherent vectors in the associated graph correspondence.
3. Establish a relative spectral-gap estimate that prevents these vectors from escaping weakly to zero. The commuting weakly mixing factor should eliminate the diffuse part of the limiting relation kernel.
4. Extract a nonzero graph summand in Z_D . Its polar parts give a coherent normalizing corner. The nonamenable-centralizer proper-support argument of [5, Theorem 3.2 and Remark 3.5] then forces that corner to have support one.

The required estimate can be isolated as a relative commutant lifting statement:

Conjecture 5.2 (Relative commuting-lifts principle). *Let $V(\Gamma_1), W(\Gamma_2) \subset \mathcal{N}_M(D)$ be homogeneous normalizing representations whose commutators belong to the amenable core A . For the Krogager–Vaes Bernoulli action, there are A -normalizer perturbations of V and W whose ranges commute.*

This is narrower than a general groupoid coaction theorem. It uses the product structure, freeness of the two factors, amenability of A , and the Bernoulli deformation simultaneously.

6 Status

The implications proved here are

$$Q \prec_M A \implies uQu^* = D \subset A \implies \begin{cases} D \text{ is Cartan in } A, \\ \sigma_g(D) \sim_u D \text{ inside } A \quad (g \in G). \end{cases}$$

The only remaining implication is the homogeneous-space coboundary equation

$$z^* \sigma_g(z) \in \mathcal{N}_A(D) a_g \quad (g \in G).$$

For $G = \mathbb{F}_\infty \times \mathbb{F}_\infty$, all one-factor relations can be solved exactly. The obstruction is entirely contained in extracting a coherent graph component for the mixed normalizer cocycle $c(g, h)$. Once that component has full support, Proposition 4.3 supplies the final homogeneous-space coboundary.

References

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